perplexity

for the Runtime Delta

Created an editable draw.io swimlane package with two pages:
Runtime Delta Flow
Enterprise source change and CDC/outbox capture
Governed MSK topic publication
Direct Kafka versus managed-relay decision
Vendor-specific consumer group and offsets
Cohere AI/agentic processing
Durable checkpoint before offset commit
Optional acknowledgment/result event
Retry, quarantine, replay, reconciliation, and lag controls

we are healthcare company and vendors (eg one of the vendor Cohere -coherehealth.com) will sync with our systems and we process delta,

based on your Swim lane diagram and overall enterprise architecture diagram.
(VALIDATE my request below before answering it - if some request is invalid let me know)

1. I need every detail for every step along the way including independent delta position, failure controls, identity & access management (OIDC, CIAM etc) - refer to your slide "vendor triggered delta consumption - kafka consumers poll the broker, the enterprise publishes deltas independently of when the vendor is ready to process them".
2. I need list of things that needs to be configured in MSK and other systems (tools, processes, functions where necessary)
3. Pros & Cons
4. SLA/SLO from us
5. Security demands from vendor systems

Your request is valid, but the runtime design needs several clarifications before it becomes an implementation-ready healthcare integration. Most importantly, distinguish **publishing enterprise changes**, **vendor ingestion**, and **successful synchronization of the vendor's business systems**—those are different responsibilities and different success measures.

The assessment below validates the architecture you described. It does **not** confirm that Cohere Health supports direct Kafka connectivity, a particular authentication mechanism, or an acknowledgment API; those capabilities require vendor confirmation.

Validate the design

Request or assumption	Assessment	Required clarification
Enterprise publishes independently of vendor readiness	**Valid**	Publication continues while the vendor is offline, subject to enterprise capacity and retention commitments. Kafka consumption does not remove records; retention controls their availability. [1]
Each vendor has an independent delta position	**Valid, with terminology correction**	Separate **source capture position**, **Kafka delivery position**, and **business synchronization position**. They must not be treated as one checkpoint.
Enterprise processes delta, while vendors synchronize their systems	**Valid**	Enterprise detects and publishes source changes; vendors apply those changes to their own systems. If vendors also return changes, that is a separate inbound flow with enterprise ETL.
A vendor's last timestamp is sufficient to resume	**Not sufficient by itself**	Use Kafka offsets for transport recovery and business versions/change identifiers for correct synchronization. Source metadata can include log positions, transaction identifiers, and ordering information. [2] [3]
Durable checkpoint before offset commit means synchronization succeeded	**Conditionally valid**	A durable inbox proves receipt, not necessarily target-system synchronization. Define which milestone the offset represents.
OIDC/CIAM authenticates every Kafka consumer directly	**Do not assume this**	Separate human login, API authentication, and Kafka workload authentication. An Entra-issued token is not automatically an MSK credential.
Optional acknowledgment is enough to offer an end-to-end synchronization SLA	**Not valid**	Without vendor business acknowledgments or equivalent telemetry, enterprise cannot substantiate when the vendor's target system became current.
The enterprise does not own vendor delta logic	**Valid ownership model**	Enterprise still owns the published contract, completeness controls, authorized access, retention, and recovery procedures—not vendor-specific transformation or synchronization state.
"Cohere AI/agentic processing" belongs in the standard runtime	**Unverified and unnecessary**	Replace this with **Vendor ingestion and business processing** unless that particular contracted workflow actually uses AI.

Working baseline: a private **MSK Provisioned cluster using Standard brokers**, with a three-Availability-Zone design, vendor-specific delivery topics, and either direct Kafka consumption or an enterprise-managed delivery adapter. This baseline avoids assuming that every configuration is available identically across Standard, Express, and Serverless offerings.

Ownership and service boundaries

Use these five swimlanes to make the architecture understandable to both Kafka architects and product owners.

Swimlane	Components	Accountable responsibility
Enterprise source/application teams	Source applications, databases, transactional outbox, source business rules	Correct source transactions, source identifiers, change semantics, approved data scope

Swimlane	Components	Accountable responsibility
Enterprise integration/data team	CDC connector, transformation service, publication pipeline, vendor projections, inbound ETL	Capture, normalize, validate, publish, reconcile, and process returned vendor changes
Enterprise MSK platform team	Brokers, networking, topic provisioning, access policies, monitoring, capacity, platform recovery	Operate the Kafka service and its access boundary
Vendor integration team	Kafka consumer or API client, durable inbox, delta processor, target-system adapter, checkpoint store	Consume independently, transform, deduplicate, synchronize, recover, and acknowledge when agreed
Shared security and operations	Identity administration, privacy/legal review, incident response, change approval, evidence retention	Approve and continuously verify the cross-company trust relationship

Important: the MSK platform team should not automatically inherit ownership of source-data correctness, enterprise ETL business rules, or vendor synchronization logic.

Independent delta positions

A production design needs **at least four distinguishable positions.**

Position	Owner	What it represents	Storage
Source capture position	Enterprise integration team	Which committed source changes have been captured	CDC connector offsets/source-log checkpoint
Published position	Enterprise integration team	Which captured events reached the governed delivery stream	Publication ledger, lineage, topic/partition/offset references
Vendor delivery position	Vendor integration team	Which Kafka records the vendor has durably accepted or processed, according to its commit policy	Consumer-group offsets and, where needed, vendor inbox
Vendor business-applied position	Vendor application team	Which changes actually reached the vendor's target business system	Vendor synchronization ledger and target transaction state

CDC systems such as Debezium expose source-log offsets and transaction metadata; that source position is different from the downstream Kafka consumer position. [2] [3]

Consumer-group isolation

Recommended naming convention:

```
Topic:
prod.vendor.cohere.authorizations.delta.v1

Consumer group:
prod.cohere.authorizations.sync.v1
```

For another vendor:

```
Topic:
prod.vendor.vendor-b.authorizations.delta.v1

Consumer group:
prod.vendor-b.authorizations.sync.v1
```

Design rules:

- Use **one group per independent processing application**, not merely one group per company.
- Instances of the same application share its group.
- A vendor's analytics application and operational synchronization application need different groups.
- Vendor A must not be authorized to use Vendor B's group.
- Grant access only to the vendor's approved topics and group namespace.

AWS's subscriber-delivery guidance illustrates dedicated delivery topics, private access, and permissions scoped to both topics and consumer groups.[4]

Commit semantics

Choose and document one of these two models.

Model	When the Kafka offset advances	Meaning	Main consequence
Apply before commit	After a successful target-system transaction	Records through the committed position have been applied under the agreed processing rules	Slow target systems can directly increase Kafka lag
Durable inbox before commit	After records are safely stored in the vendor's durable inbox	Records have been accepted for processing	Kafka lag can be low while business synchronization is far behind

Recommendation: allow a durable inbox for vendors with complex or slow target systems, but require separate **inbox backlog** and **business-applied freshness** measurements. Do not present Kafka offset advancement as proof of business synchronization.

Detailed runtime sequence

Onboarding prerequisites

Before runtime access, establish:

1. Approved use case, data owner, and permitted use.
2. Vendor's verified integration capabilities.
3. Selected delivery mode: direct Kafka or adapter.

4. Data contract and schema.
5. Initial-load and incremental-change strategy.
6. Retention and resynchronization policy.
7. Authentication and authorization design.
8. Security/privacy agreements and operational contacts.
9. Certification tests and production approval.

Where the vendor acts as a HIPAA business associate, the required written assurances must cover permitted uses and appropriate safeguards; do not assume that a technical connection alone authorizes PHI disclosure. [5]

Runtime steps and controls

Step	Owner and components	Required behavior	Failure controls
1. Commit a business change	Enterprise source application and database	Commit the authoritative business transaction; assign an entity identifier and version/change identity	Avoid an uncontrolled database-write-plus-Kafka-write dual-write
2. Capture the change	CDC connector or transactional outbox pipeline	Capture committed changes, including agreed update/delete semantics	Monitor connector health, source-log continuity, credentials, and capture lag
3. Normalize the event	Enterprise transformation service	Convert raw changes into the approved business event; preserve source lineage	Reject invalid mappings; quarantine malformed events without silently losing them
4. Enforce vendor entitlement	Vendor projection service	Select permitted entities and fields before external publication	Fail closed if entitlement cannot be evaluated
5. Validate the contract	Producer serializer, schema registry, validation rules	Validate required fields, schema version, operation, business version, and envelope	Block incompatible publication; alert contract owners
6. Publish to MSK	Enterprise producer	Publish with an appropriate entity key and durability settings	Retry transient failures; preserve event identity; alert on sustained publication failure
7. Retain the delivery stream	MSK platform	Keep events for the contracted recovery window	Monitor capacity, retention configuration, and oldest available offsets
8. Vendor connects	Vendor workload and enterprise network/access controls	Establish private connectivity and authenticate the approved workload	Deny unauthorized identities, topics, groups, and environments
9. Vendor resumes	Vendor consumer and checkpoint logic	Resume from its own group position or an explicitly approved recovery position	Fail visibly on missing/expired positions; do not silently skip history

Step	Owner and components	Required behavior	Failure controls
10. Vendor polls	Vendor Kafka client	Fetch records within agreed throughput and concurrency limits	Backoff, bounded buffering, partition pause/resume, and rebalance handling
11. Vendor validates/deduplicates	Vendor inbox or processing service	Validate schema, event identity, authorization context, and business version	Detect duplicates and stale updates; quarantine invalid records
12. Vendor applies delta	Vendor transformation and target adapter	Upsert/delete/correct target records according to the contract	Use idempotent application and a durable processing ledger
13. Vendor advances checkpoints	Vendor consumer and ledger	Advance only through successfully handled contiguous records per partition	Do not commit beyond a failed record unless it has been durably transferred to an approved recovery path
14. Vendor acknowledges	Optional status producer/API	Distinguish received, applied, rejected, and retrying	Make acknowledgments idempotent and correlate them to the original event
15. Reconcile and recover	Enterprise integration and vendor teams	Compare agreed publication scope against vendor application evidence	Controlled replay, repair extract, or new baseline
16. Process returned changes	Enterprise inbound adapter and ETL	If bidirectional, validate and merge vendor changes into enterprise targets	Separate inbound identity, checkpoints, deduplication, conflict rules, and reconciliation

A transactional outbox can persist the event within the business database transaction and then use CDC to relay it asynchronously. AWS recommends producer durability settings such as `acks=all` and idempotence, with replication and minimum in-sync replicas configured to tolerate broker unavailability.[6] [7]

Event contract

Recommended event envelope—not a claim about Cohere Health's existing interface:

```
{
  "event_id": "globally-unique-change-identifier",
  "event_type": "AuthorizationUpdated",
  "schema_version": "1",
  "entity_id": "opaque-business-identifier",
  "entity_version": "source-controlled-version",
  "operation": "UPSERT",
  "source_system": "enterprise-authorizations",
  "source_commit_time": "UTC timestamp",
  "published_at": "UTC timestamp",
  "correlation_id": "workflow-identifier",
  "causation_id": "prior-event-or-command-identifier",
  "publication_run_id": "optional-daily-run-identifier",
```

```
  "payload": {}
}
```

Specify these semantics separately:

- Whether events contain complete entity state or only changed fields.
- How deletes, cancellations, reversals, and corrections are represented.
- Whether versions are monotonic per entity.
- How related entities and cross-record transactions are handled.
- Which fields may contain PHI.
- How schema evolution and historical replay work.
- Whether a daily publication is a **snapshot** or a **delta batch**.

Do not place patient names, diagnoses, or other unnecessary PHI in topic names, group names, metric dimensions, or diagnostic labels.

Initial synchronization and daily pulls

Initial synchronization

Recommended sequence:

1. Agree the authorized baseline population.
2. Establish a consistent source cut or connector-supported snapshot boundary.
3. Deliver the baseline with an identifiable snapshot/run ID.
4. Preserve changes that occur during baseline delivery.
5. Apply the baseline and subsequent changes without gaps.
6. Reconcile before declaring the vendor current.

Debezium's database connectors document snapshot procedures that record source positions and then transition into change streaming; the exact procedure depends on the source database and connector.[2] [3]

Do not define initial synchronization as "read the database, then start consuming latest." That leaves an undefined gap between the two activities.

Daily publication pattern

Your earlier Pattern 4 remains valid, with this responsibility split:

Enterprise provides	Vendor owns
A documented daily publication schedule	When to run its consumer
Complete publication for the defined scope	Last successful processing position

Enterprise provides	Vendor owns
Run ID, cutoff, schema, and completion evidence	Delta calculation against its own system
Retained data for the agreed recovery window	Target transformation, deduplication, and application
Defined replay/resnapshot procedure	Detecting missed runs and requesting recovery

Recommended daily publication controls:

- A run manifest identifying snapshot versus delta.
- Source cutoff and publication completion time.
- Expected records or reconciliation aggregates, where meaningful.
- Partition-aware completion boundaries.
- Explicit delete semantics.
- A defined response to incomplete or superseded runs.

A marker in one partition should not be treated as proof that every other partition's records have been processed. Design completion around a manifest and partition boundaries, not a single globally assumed end marker.

For full snapshots, absence means deletion **only if the contract explicitly defines it that way**.

Identity and access design

Separate identity planes

Plane	Recommended approach	What it authorizes
Enterprise workforce	Microsoft Entra workforce federation, MFA, approved administrative roles	Platform administration and support tools
Vendor human users	B2B federation or CIAM, depending on the portal's audience	Onboarding portal, documentation, requests, support
Vendor Kafka workload	An explicitly selected MSK-supported workload authentication path	Kafka connections and permitted Kafka operations
Vendor API workload	OAuth client credentials and/or mTLS, according to gateway design	Adapter, acknowledgment, or management APIs

OIDC, CIAM, and Kafka authentication are not interchangeable controls. Treat the following as the recommended identity blueprint; the exact Entra-to-AWS federation and client configuration must be validated against your selected runtime.

Direct Kafka reference path

Recommended AWS-native pattern:

```
Vendor application workload
        |
        | obtains short-lived AWS credentials
        | through approved role/federation arrangement
        v
MSK-compatible IAM authentication client
        |
        | authenticated encrypted connection
        v
Enterprise MSK cluster
        |
        | topic and consumer-group authorization
        v
Vendor-approved delivery topic
```

For cross-company role access:

- Bind trust to the approved vendor role or federation subject.
- Avoid long-lived shared access keys.
- Use an external ID where appropriate for third-party role assumption.
- Restrict production and nonproduction separately.
- Record attributable sessions.
- Test renewal, revocation, and incident containment.
- Do not give vendor workloads platform-administrator permissions.

AWS's delivery guidance scopes IAM Kafka access to a subscription's topic and consumer group. [4]

Minimum access scope

For a **read-only vendor consumer**, the intended permission scope is:

Resource	Intended permissions
Approved cluster	Connect
Approved delivery topic	Describe and read
Approved consumer group	Describe and participate/advance offsets
Schema distribution mechanism	Read only the schemas needed by the subscription
Other vendor topics/groups	No access
Topic creation, deletion, configuration	No access
Enterprise raw CDC topics	No access

For IAM-based MSK authorization, the corresponding data-plane actions commonly include `kafka-cluster:Connect`, `DescribeTopic`, `ReadData`, `DescribeGroup`, and `AlterGroup`; AWS documents this topic/group-scoped subscriber pattern.[4]

If a vendor returns acknowledgments or results, grant a **separate narrowly scoped write path**. Read access must not imply broad write access.

API/relay identity path

For vendors that cannot operate a supported Kafka client:

```
Vendor workload
      |
      | API token and/or client certificate
      v
Enterprise API gateway / delivery service
      |
      | enterprise workload identity
      v
Enterprise consumer / durable delivery state
      |
      v
MSK
```

The API contract must distinguish:

- **Fetch:** records were returned.
- **Accept:** records were durably accepted.
- **Apply:** business synchronization succeeded.

If the enterprise builds a pull API with per-vendor continuation state, it now owns that **API delivery cursor**. That is additional state beyond direct vendor-owned Kafka consumption.

Configuration checklist

MSK platform

The settings below are a **proposed baseline**, not a universal prescription. Verify support against the selected MSK deployment, broker type, Kafka version, and actual workload.

Configuration area	What to configure	Ownership
Deployment	Region, broker type, Kafka version, Availability Zones, environment separation	MSK platform
Capacity	Brokers, storage, partitions, throughput headroom, recovery headroom	MSK platform
Replication	For the stated three-AZ baseline, replication factor 3 and `min.insync.replicas=2`	MSK platform
Producer durability	`acks=all`, explicitly reviewed idempotence and retry behavior	Integration team

Configuration area	What to configure	Ownership
Encryption	At-rest encryption and authenticated encrypted client connections	MSK/security
Authentication	Approved method; prohibit unauthenticated vendor access	MSK/security
Network	Private connectivity, security groups, DNS, routes, endpoints	Network/MSK
Cross-account access	Cluster policy, role trust, managed connection permissions where applicable	MSK/security/vendor
Topic management	Explicit topic creation, naming, ownership, permitted configurations	MSK platform
Retention	Contracted retention window and storage-size limits	MSK/product owner
Cleanup	Select deletion/compaction according to event-history versus latest-state requirements	Integration/MSK
Group offsets	Offset retention appropriate for inactive/scheduled vendors	MSK platform
Tenant limits	Throughput, connection, and request protections	MSK platform
Monitoring	Broker, partition, capacity, authentication, consumer, and publication signals	MSK/operations
Recovery	Recovery topology, restoration procedure, failover and replay tests	MSK/integration
Change management	Version/configuration changes, maintenance testing, rollback	MSK platform

The replication/producer baseline follows AWS's documented durability guidance. [6]

Retention has two independent dimensions: retained topic data and retained consumer-group offsets. AWS documents that inactive groups can lose their stored offsets after the configured offset-retention period; retaining topic data alone does not guarantee the group's resume position remains available. [8]

Also account for time **and size** retention constraints: a size limit can undermine a promised time-based recovery window if capacity is insufficient. [1]

Connectivity considerations

For managed MSK multi-VPC private connectivity, verify its specific deployment and Region constraints rather than assuming every PrivateLink capability applies identically. AWS's subscriber guidance describes same-Region connectivity and client-subnet/AZ requirements. [4]

For vendors outside that supported pattern, assess an approved network design or enterprise adapter instead of exposing brokers publicly by default.

Other enterprise systems

System/component	Required configuration
Source database	CDC/logging prerequisites, capture account, source-log retention, supported delete behavior
CDC service	Included tables/columns, snapshot mode, checkpoint store, retries, source monitoring
Outbox publisher	Transactional event creation, immutable event ID, cleanup and replay rules

System/component	Required configuration
Transformation runtime	Mappings, version rules, validation, entitlement projections, quarantine
Schema management	Compatibility policy, approval workflow, vendor schema access/distribution
Publication ledger	Source/run lineage, publication failures, completeness evidence
Secrets/key services	Key ownership, secret/certificate rotation, access restrictions
Identity service	Human versus workload identities, federation trust, lifecycle and revocation
Observability	Dashboards, alerts, synthetic checks, escalation routes, PHI-safe logging
Recovery storage	Authorized baseline/repair extracts, retention, encryption, controlled access
Inbound ETL	Vendor extraction, staging, transformation, target merge, checkpoints, reconciliation
Service management	Subscription catalog, operational contacts, incidents, changes, offboarding evidence

Choose a CDC implementation rather than assuming that all tools have identical behavior. For example, AWS DMS supports Kafka/MSK targets with object mapping and partition-key configuration; connector capabilities still need to be matched to the source and required event semantics.[9]

Vendor consumer configuration

Setting/control	Recommended requirement
`group.id`	Approved, stable application-specific group
Automatic offset commits	Disable when implementing explicit durable-processing semantics
Missing-offset behavior	Fail visibly and invoke the agreed recovery policy; do not silently choose latest
Poll limits	Align record count and bytes with actual processing capacity
Poll interval/rebalance behavior	Test slow processing, failures, and partition reassignment
Checkpointing	Advance only contiguous successfully handled positions
Parallel processing	Preserve the required entity/partition ordering
Buffering	Bounded memory and durable spill/inbox where needed
Idempotency	Stable event identity plus target-side application protection
Version handling	Reject stale changes and define equal-version conflict behavior
Credentials	Short-lived credentials or managed secret/certificate lifecycle
Client compatibility	Certified against the selected MSK authentication and broker version
Telemetry	Delivery lag, inbox backlog, applied freshness, failures, retries, quarantine
Shutdown	Stop safely, finish or abandon work explicitly, and prevent stale commits

Failure controls

Failure	Required response	Primary owner
Source capture stops	Alert; preserve source logs; recover from capture checkpoint	Enterprise integration
Source logs expire before capture	Stop claiming continuity; perform controlled rebaseline/repair	Enterprise integration/source team
Publication fails	Retry without losing event identity; reconcile missing publication	Enterprise integration
Vendor loses connectivity	Resume from retained position after recovery	Vendor/network teams
Vendor crashes before applying	Reprocess safely from its durable state	Vendor
Vendor applies but crashes before Kafka commit	Expect replay; deduplicate/idempotently reapply	Vendor
Vendor commits before durable receipt/application	Treat as a correctness defect; repair from evidence or replay	Vendor
Vendor inbox fills	Backpressure and alert; stop accepting beyond safe capacity	Vendor
Poison event	Durably quarantine or pause according to ordering/clinical policy	Vendor, with enterprise contract owner
Out-of-order/stale update	Apply entity-version rules; do not overwrite newer state blindly	Vendor
Consumer rebalance	Stop stale processing/commits; transfer ownership safely	Vendor
Retained history no longer covers vendor position	Controlled snapshot/repair; never silent skipping	Shared
Unauthorized access	Deny, alert, investigate, and execute containment runbook	Security/MSK
Schema incompatibility	Reject change or quarantine; coordinate contract remediation	Enterprise/vendor
Returned vendor event echoes outbound event	Use origin, causation, and version/conflict controls	Shared integration teams

Quarantine and replay rules

A dead-letter or quarantine queue is **not** a substitute for successful synchronization.

Require:

- A durable record of the original event identity and failure reason.
- Restricted PHI access.
- A named remediation owner.
- A resolution deadline based on business impact.

- Idempotent reprocessing.
- Audit evidence linking repair to the original event.
- Explicit rules on whether subsequent updates for the same entity may proceed.

For a clinically important sequence, proceeding past a failed event may be unacceptable. Decide that per business workflow rather than letting a generic connector configuration decide it.

Exactly-once boundary

Do not promise **end-to-end exactly-once synchronization** merely because the producer uses idempotence.

The proposed contract should normally say:

> **At-least-once delivery, with idempotent business application and auditable recovery.**

AWS describes producer idempotence as reducing duplicate-message likelihood; that is not evidence of an atomic transaction spanning the enterprise source, Kafka, and the vendor's target database.[6]

Enterprise inbound ETL

Your earlier correction remains important: **pulling vendor data into Kafka is not the same as synchronizing enterprise systems.**

Recommended inbound path:

```
Vendor API / export / approved event stream
                  |
                  v
Enterprise extractor and extraction checkpoint
                  |
                  v
Raw landing / inbound Kafka topic
                  |
                  v
Enterprise validation and transformation
                  |
                  v
Deduplication and business-version/conflict checks
                  |
                  v
Enterprise target merge / approved application API
                  |
                  v
ETL applied checkpoint and reconciliation
```

Keep these independent:

- Vendor extraction position.

- Inbound Kafka position.
- Enterprise target-applied position.
- Business acknowledgment/status.

Do not acknowledge “enterprise synchronized” merely because the vendor payload landed in an inbound topic.

For bidirectional flows, define the authoritative system **per entity or field**. Without that rule, retries and returned updates can create loops or overwrite legitimate changes.

Pros and cons

Direct Kafka versus adapter

Dimension	Direct vendor Kafka	Enterprise-managed adapter
Independence	Vendor controls consumption and recovery position	Depends on adapter API and delivery-state design
Vendor prerequisites	Kafka client, supported authentication, network capability	Usually a simpler API integration
Enterprise runtime burden	Lower delivery-application burden	Enterprise operates another application and its state
Security surface	Vendor reaches a tightly scoped Kafka boundary	Vendor reaches an API boundary; brokers remain internal
Delta ownership	Vendor can own both delivery and business positions	Enterprise often owns API cursor; vendor owns business delta
Backpressure	Vendor consumption rate directly controls its backlog	Adapter must handle buffering, throttling, and acknowledgments
Observability	Kafka position plus vendor-reported application status	Better request-level visibility, but still needs business status
Isolation	Requires careful topic/group policies and capacity controls	Application-level entitlement and throttling controls
Cost	Connectivity, broker capacity, retention, vendor client operations	Those costs plus adapter compute, state, and support
Best fit	Technically mature vendor with sustained event consumption	Vendor unable or unwilling to operate Kafka securely

Overall pattern trade-offs

Advantages

- Enterprise publishing is decoupled from vendor availability.
- Each vendor can process at its own pace.
- Retained history enables controlled replay.

- A common event contract supports multiple approved consumers.
- Vendor business synchronization logic stays with the vendor.

Disadvantages

- Eventual consistency must be acceptable to the business.
- Retention limits the offline recovery window.
- Duplicate handling, deletes, corrections, and ordering require explicit design.
- Kafka delivery position alone does not prove business synchronization.
- More vendors increase entitlement, schema, capacity, and support complexity.
- Daily snapshots can shift substantial comparison and reconciliation work to vendors.

These trade-offs are architectural assessments of the proposed design—not measured outcomes for your environment.

SLA and SLO framework

Do not offer one undifferentiated "Kafka synchronization SLA." Publish separate objectives for the parts your company controls, with a joint objective only where vendor evidence supports it.

Define the clocks

Milestone	Meaning
T0	Source transaction committed
T1	Valid event available on the authorized delivery topic
T2	Vendor durably accepted the event
T3	Vendor target system successfully applied the event
T4	Enterprise received the agreed business acknowledgment

Measure:

- **Enterprise publication latency:** T1 minus T0.
- **Vendor delivery latency:** T2 minus T1.
- **Vendor processing latency:** T3 minus T2.
- **End-to-end synchronization latency:** T3 minus T0.

Cross-company measurements need agreed clock handling, event identifiers, and trusted status evidence.

Proposed starting objectives

These are negotiation candidates—not AWS guarantees, existing commitments, or measured performance. Benchmark and approve them before including them in a contract.

Service measure	Illustrative starting objective	Accountability
Authorized delivery-path availability	99.9% per calendar month	Enterprise, within explicitly defined boundary
Streaming publication freshness	95% within 5 minutes; 99% within 15 minutes of source commit	Enterprise source/integration/platform teams
Daily publication	Complete by the contracted cutoff, with a completion manifest	Enterprise
Online replay window	At least 14 days, subject to approved PHI retention and capacity	Enterprise
Vendor consumption cadence	Within the agreed interval for its service tier	Vendor
Vendor business-applied freshness	Agreed percentile/deadline after publication	Vendor; joint reporting if enterprise receives evidence
Critical incident response	Acknowledgment within 15 minutes for a contracted 24×7 tier	Respective on-call teams
Recovery time and recovery point	Values established by tested recovery design before signing	Enterprise and vendor for their respective components

For a daily-only service, replace streaming freshness targets with **publication cutoff, completeness, and recovery** objectives; do not sell a continuous-freshness promise for a daily feed.

Contract definitions

The SLA must specify:

- Measurement point and denominator.
- Service hours and support coverage.
- Eligible versus invalid source transactions.
- Planned maintenance treatment.
- Vendor/network dependency boundaries.
- Maximum supported volume, record size, and connection profile.
- Retention and missed-window recovery.
- Publication completeness versus target application success.
- Incident notification, escalation, and service remedies.
- Recovery dependencies and tested RTO/RPO.
- What happens when a vendor consumes outside the supported envelope.

The AWS service SLA is an upstream dependency, not automatically your customer-facing SLA. Verify its current terms for the selected MSK offering separately.

Retention sizing

Use this planning relationship:

> **Recovery window ≥ maximum vendor outage + detection delay + repair time + catch-up duration + safety margin**

Also validate that the vendor can consume faster than new events arrive during recovery; otherwise it cannot clear its backlog.

Do not interpret the window as a legal requirement to retain PHI. Privacy, legal, capacity, and business recovery requirements must be approved together.

Vendor security requirements

Treat these as **recommended enterprise admission requirements**, with legal obligations determined by the actual relationship and jurisdiction.

Legal and data use

- Establish the vendor's legal role and permitted purpose.
- Execute a BAA where required.
- Restrict use to the contracted healthcare function.
- Approve subcontractors, hosting locations, and onward disclosures.
- Define retention, return/deletion, and termination handling.
- Prohibit unapproved secondary use, including AI/model training.
- Define security-incident and breach-reporting obligations with counsel.

HHS explains that a covered entity's business-associate arrangements must specify permitted uses and require appropriate safeguards; business-associate access is not permission for independent use of the information. [5]

Workload and infrastructure security

Require:

- Dedicated production workload identities.
- No shared human accounts or embedded long-lived credentials.
- Supported authentication libraries and credential renewal.
- Encrypted transport and storage.
- Secrets and certificate lifecycle management.
- Hardened, supported runtime images.

- Vulnerability and dependency management.
- Environment and tenant separation.
- Restricted administrative access with MFA.
- Controlled egress and approved subprocessors.
- Secure backup and restoration procedures.

A SOC 2 report or similar assurance package may support due diligence; it should not be treated as proof that this particular data flow is correctly configured.

Data handling and correctness

Require:

- Only approved PHI fields and populations.
- No PHI in unrestricted logs, traces, tickets, or metrics.
- Idempotent application and duplicate detection.
- Correct handling of deletes, corrections, and stale versions.
- Durable checkpoint/inbox protection.
- Independent monitoring of delivery and business-applied positions.
- Reconciliation and controlled replay.
- Evidence of target-system isolation.
- Approved retention and deletion enforcement.

Monitoring and incident response

Require:

- Named operational and security contacts.
- Coverage appropriate to the contracted service tier.
- Alerts for consumer failure, credential failure, backlog growth, and impending retention loss.
- Prompt containment and evidence preservation.
- Agreed incident notification timelines.
- Investigation cooperation and access to relevant audit evidence.
- Tested business continuity and restoration.
- Controlled reauthorization after a security event.

Do not assume that removing a portal user terminates an already running workload. The containment runbook must address workload credentials, role trust, Kafka authorization, network access, and active sessions/connections as applicable.

Certification before production

Use these as release gates:

Test	Required result
Vendor A attempts Vendor B's topic/group	Access denied
Vendor attempts unapproved raw CDC access	Access denied
Workload credentials expire or rotate	Safe renewal or explicit failure; no silent loss
Vendor is offline within retention	Complete catch-up without gaps
Vendor exceeds retention	Controlled rebaseline; no silent jump to latest
Consumer crashes after target commit	Replay does not duplicate business effect
Consumer rebalances during processing	No stale commits or missing records
Duplicate, late, and stale events arrive	Correct deterministic handling
Delete/cancellation/correction arrives	Agreed target behavior
Poison event arrives	Durable, auditable failure handling
Schema changes	Compatibility enforced and vendor behavior verified
Daily publication is incomplete	Vendor does not declare synchronization complete
Vendor target is slow	Backpressure works and applied-lag alert fires
Enterprise inbound ETL fails	Landing success is not reported as target synchronization
Vendor is offboarded	Access revoked and data disposition evidenced

The key service boundary should be explicit: your company guarantees the authorized, contract-compliant publication and recovery service; the vendor owns its consumption position and target synchronization. An end-to-end synchronization commitment requires a separately agreed vendor acknowledgment and reconciliation contract.

⁂

1. https://docs.amazonaws.cn/msk/latest/developerguide/bestpractices.html
2. https://debezium.io/documentation/reference/stable/connectors/sqlserver.html
3. https://debezium.io/documentation/reference/stable/connectors/postgresql.html
4. https://docs.aws.amazon.com/it_it/guidance/latest/connected-mobility-on-aws/tpd-realtime-b2b.md
5. https://www.hhs.gov/guidance/document/bas
6. https://docs.aws.amazon.com/msk/latest/developerguide/version-upgrades-best-practices.html
7. https://debezium.io/documentation/reference/stable/integrations/outbox.html
8. https://docs.aws.amazon.com/en_us/msk/latest/developerguide/msk-configuration-express-read-write.html
9. https://docs.aws.amazon.com/dms/latest/userguide/CHAP_Target.Kafka.html
10. https://agenticindex.io/vendors/cohere-health

11. https://www.coherehealth.com/utilization-management/policy-studio
12. https://www.coherehealth.com/utilization-management-suite
13. https://www.coherehealth.com/blog/cms-0057-f-apis-are-only-the-tip-of-the-iceberg
14. https://www.coherehealth.com/blog/two-provisions-to-watch-in-the-cms-prior-authorization-rule
15. https://www.coherehealth.com/news/new-prior-auth-legislation
16. https://docs.aws.amazon.com/msk/latest/developerguide/kafka-actions.md
17. https://docs.aws.amazon.com/msk/latest/developerguide/security-iam-awsmanpol-AWSMSKReplicatorExecutionRole.md
18. https://docs.aws.amazon.com/msk/latest/developerguide/getting-started.md
19. https://docs.aws.amazon.com/msk/latest/developerguide/msk-encryption.html
20. https://docs.aws.amazon.com/msk/latest/developerguide/msk-working-with-encryption.html
21. https://docs.aws.amazon.com/us_en/eks/latest/userguide/authenticate-oidc-identity-provider.html
22. https://docs.aws.amazon.com/msk/latest/developerguide/consumer-lag.html
23. https://docs.aws.amazon.com/msk/latest/developerguide/msk-authentication-cluster.html
24. https://docs.aws.amazon.com/msk/latest/developerguide/msk-data-delivery-s3-security-rest.html
25. https://docs.aws.amazon.com/msk/latest/developerguide/msk-working-with-encryption-test-tls.html
26. https://docs.aws.amazon.com/msk/latest/developerguide/msk-replicator-encryption.html
27. https://docs.aws.amazon.com/msk/latest/developerguide/msk-password-tutorial.html
28. https://www.hhs.gov/sites/default/files/breach-report-to-congress-2019.pdf
29. https://www.hhs.gov/hipaa/for-professionals/faq/smaller-providers-and-businesses/index.html
30. https://www.hhs.gov/hipaa/for-professionals/faq/how-may-hipaas-minimum-necessary-standard-apply-to-electronic-information/index.html
31. https://www.hhs.gov/sites/default/files/emergencysituations.pdf
32. https://www.coherehealth.com/ai-headquarters
33. https://www.hhs.gov/hipaa/for-professionals/special-topics/emergency-preparedness/hipaa-limited-waiver-2026-phe-typhoon-bavi.html
34. https://www.hhs.gov/hipaa/for-professionals/special-topics/emergency-preparedness/disclosures-subject-to-minimum-necessary-standard/index.html
35. https://www.hhs.gov/guidance/sites/default/files/hhs-guidance-documents/CMS/mln909001_2025_05_hipaa_basics_final.pdf
36. https://www.hhs.gov/sites/default/files/ocr/privacy/hipaa/understanding/coveredentities/minimumnecessary.pdf
37. https://www.hhs.gov/hipaa/for-professionals/faq/208/wont-minium-necessary-restriction-impede-delivery/index.html
38. https://www.hhs.gov/hipaa/for-professionals/breach-notification/index.html
39. https://www.hhs.gov/hipaa/for-professionals/privacy/guidance/minimum-necessary-requirement/index.html
40. https://www.hhs.gov/hipaa/for-professionals/faq/business-associates/index.html
41. https://kafka.apache.org/41/javadoc/constant-values.html
42. https://docs.aws.amazon.com/msk/latest/developerguide/msk-default-configuration.md
43. https://docs.aws.amazon.com/en_en/IAM/latest/UserGuide/id_federation_common_scenarios.html

44. https://docs.aws.amazon.com/devopsagent/latest/userguide/aws-devops-agent-security-setting-up-external-identity-provider-idp-authentication.html
45. https://docs.aws.amazon.com/ja_jp/devopsagent/latest/userguide/aws-devops-agent-security-setting-up-external-identity-provider-idp-authentication.md
46. https://learn.microsoft.com/en-us/entra/external-id/customers/how-to-custom-oidc-federation-customers
47. https://techcommunity.microsoft.com/blog/microsoft-entra-blog/customize-authentication-experiences-and-url-domains-for-external-apps/3627347
48. https://docs.aws.amazon.com/en_en/IAM/latest/UserGuide/id_roles_providers_oidc.html
49. https://docs.aws.amazon.com/msk/latest/developerguide/bestpractices.html
50. https://devblogs.microsoft.com/identity/openid-connect-external-identity-provider-support-ga/
51. https://kafka.apache.org/41/javadoc/org/apache/kafka/clients/consumer/ConsumerConfig.html
52. https://learn.microsoft.com/nb-no/ entra/external-id/customers/how-to-custom-oidc-federation-customers
53. https://docs.aws.amazon.com/msk/latest/developerguide/msk-configuration-express-read-only.md
54. https://d1.awsstatic.com/legal/msk-sla/Amazon_Managed_Streaming-Apache_Kafka_SLA_Korean_2022-05-19.pdf
55. https://aws.amazon.com/marketplace/pp/prodview-kyrly5yzbhe7s
56. https://docs.aws.amazon.com/msk/latest/developerguide/msk-data-delivery-iceberg-cloudtrail.md
57. https://docs.aws.amazon.com/msk/latest/developerguide/msk-replicator-bp-quotas.html
58. https://docs.aws.amazon.com/msk/latest/developerguide/limits.html
59. https://docs.aws.amazon.com/msk/latest/developerguide/msk-logging.html
60. https://docs.aws.amazon.com/msk/latest/developerguide/msk-data-delivery-s3-cloudtrail.html
61. https://docs.aws.amazon.com/msk/latest/developerguide/create-client-iam-role.html
62. https://docs.aws.amazon.com/msk/latest/developerguide/create-iam-role.html
63. https://docs.aws.amazon.com/msk/latest/developerguide/msk-replicator-create-iam-perms.html
64. https://docs.aws.amazon.com/msk/latest/developerguide/msk-replicator-iam-ref.html
65. https://docs.aws.amazon.com/IAM/latest/UserGuide/id_roles_providers.html
66. https://docs.aws.amazon.com/msk/latest/developerguide/security_iam_troubleshoot.html
67. https://docs.aws.amazon.com/msk/latest/developerguide/data-protection.html
68. https://docs.aws.amazon.com/msk/latest/developerguide/logging-API-using-cloudtrail.html
69. https://kafka.apache.org/26/javadoc/org/apache/kafka/clients/consumer/KafkaConsumer.html
70. https://kafka.apache.org/31/javadoc/org/apache/kafka/clients/consumer/KafkaConsumer.html
71. https://kafka.apache.org/35/javadoc/org/apache/kafka/clients/consumer/KafkaConsumer.html
72. https://kafka.apache.org/27/javadoc/org/apache/kafka/clients/consumer/KafkaConsumer.html
73. https://kafka.apache.org/37/javadoc/org/apache/kafka/clients/consumer/KafkaConsumer.html
74. https://kafka.apache.org/34/javadoc/org/apache/kafka/clients/consumer/KafkaConsumer.html
75. https://kafka.apache.org/39/javadoc/org/apache/kafka/clients/consumer/KafkaConsumer.html
76. https://www.coherehealth.com/platform
77. https://kafka.apache.org/38/javadoc/org/apache/kafka/clients/consumer/KafkaConsumer.html
78. https://kafka.apache.org/30/javadoc/org/apache/kafka/clients/consumer/KafkaConsumer.html

79. https://kafka.apache.org/33/javadoc/org/apache/kafka/clients/consumer/KafkaConsumer.html
80. https://kafka.apache.org/36/javadoc/org/apache/kafka/clients/consumer/KafkaConsumer.html
81. https://kafka.apache.org/39/javadoc/org/apache/kafka/clients/consumer/Consumer.html
82. https://kafka.apache.org/33/javadoc/org/apache/kafka/clients/consumer/Consumer.html
83. https://kafka.apache.org/30/javadoc/org/apache/kafka/clients/consumer/Consumer.html
84. https://docs.aws.amazon.com/glue/latest/dg/monitor-continuations.html
85. https://docs.aws.amazon.com/msk/latest/developerguide/msk-guidelines-tiered-storage-topic-level-config.md
86. https://docs.aws.amazon.com/msk/latest/developerguide/serverless-config.html
87. https://aws.amazon.com/blogs/machine-learning/secure-ai-agents-with-amazon-bedrock-agentcore-identity-on-amazon-ecs/
88. https://docs.aws.amazon.com/msk/latest/developerguide/msk-tiered-storage-retention-rules.html
89. https://docs.aws.amazon.com/msk/latest/developerguide/serverless-monitoring.html
90. https://docs.aws.amazon.com/msk/latest/developerguide/metrics-details.html
91. https://docs.aws.amazon.com/msk/latest/developerguide/bestpractices-kafka-client.html
92. https://docs.aws.amazon.com/apigateway/latest/developerguide/http-api-jwt-authorizer.html
93. https://docs.aws.amazon.com/msk/latest/developerguide/msk-configuration-express-read-write.html
94. https://docs.aws.amazon.com/IAM/latest/UserGuide/id_roles_common-scenarios_third-party.html
95. https://docs.aws.amazon.com/msk/latest/developerguide/bestpractices-express.html
96. https://docs.aws.amazon.com/msk/latest/developerguide/msk-configuration-properties.html
97. https://docs.aws.amazon.com/msk/latest/developerguide/msk-password-howitworks.md
98. https://www.coherehealth.com/utilization-management/api-based
99. https://docs.aws.amazon.com/msk/latest/developerguide/configure-clients-for-iam-access-control.md
100. https://docs.aws.amazon.com/msk/latest/developerguide/create-cluster-console.md
101. https://docs.aws.amazon.com/msk/latest/developerguide/msk-update-security.html
102. https://docs.aws.amazon.com/msk/latest/developerguide/mvpc-cross-account-update-authschemes.md
103. https://docs.aws.amazon.com/msk/latest/developerguide/msk-password-howitworks.html
104. https://docs.aws.amazon.com/msk/latest/developerguide/msk-data-delivery-s3-cloudtrail.md
105. https://docs.aws.amazon.com/msk/latest/developerguide/msk-data-delivery-iceberg-logging.html
106. https://www.coherehealth.com/news/ai-machine-learning-reforming-prior-authorization
107. https://docs.aws.amazon.com/msk/latest/developerguide/msk-data-delivery-iceberg-bp-security.html
108. https://www.hcinnovationgroup.com/interoperability-hie/application-programming-interfaces-apis/article/55343388/qa-cohere-healths-matt-parker-on-meeting-cms-prior-authorization-api-requirements
109. https://www.coherehealth.com/glossary
110. https://docs.aws.amazon.com/pdfs/msk/latest/developerguide/MSKDevGuide.pdf
111. https://docs.aws.amazon.com/en_us/msk/latest/developerguide/logging-using-cloudtrail.html
112. https://factorhouse.io/resources/kafka/tools/kafka-audit-logging-tools/
113. https://docs.aws.amazon.com/it_it/whitepapers/latest/amazon-msk-migration-guide/amazon-managed-streaming-for-apache-kafka-amazon-msk.html

114. https://github.com/api-evangelist/cohere-health/blob/main/README.md
115. https://aws.amazon.com/msk/faqs/
116. https://docs.aws.amazon.com/zh_tw/msk/latest/developerguide/logging-API-using-cloudtrail.md
117. https://d1.awsstatic.com/legal/msk-sla/Amazon_Managed_Streaming-Apache_Kafka_SLA_Traditional Chinese_2022-05-19.pdf
118. https://d1.awsstatic.com/legal/msk-sla/Amazon_Managed_Streaming-Apache_Kafka_SLA_German_2022-05-19.pdf
119. https://aws.amazon.com/ru/msk/features/
120. https://aws.amazon.com/blogs/big-data/migrate-an-oauth-2-0-authenticated-apache-kafka-cluster-to-amazon-msk-with-msk-replicator/
121. https://resources.coherehealth.com/hubfs/Whitepapers/AHIP Considerations Guide WP/CH-AHIP Considerations Guide.pdf
122. https://aws.amazon.com/legal/service-level-agreements/
123. https://docs.aws.amazon.com/msk/latest/developerguide/aws-access-mult-vpc.md
124. https://docs.aws.amazon.com/msk/latest/developerguide/msk-replicator-metadata-acl.html
125. https://docs.aws.amazon.com/msk/latest/developerguide/public-access.md
126. https://docs.aws.amazon.com/msk/latest/developerguide/msk-acls.html
127. https://docs.aws.amazon.com/msk/latest/developerguide/msk-replicator-perms-acl-replication.html
128. https://www.hhs.gov/hipaa/for-professionals/faq/may-a-hipaa-covered-entity-or-business-associate-use-cloud-service-to-store-or-process-ephi/index.html
129. https://aws.amazon.com/compliance/hipaa-eligible-services-reference/
130. https://aws.amazon.com/compliance/services-in-scope/HIPAA_BAA/
131. https://aws.amazon.com/messaging/sla/?did=sla_card
132. https://www.hhs.gov/hipaa/for-professionals/faq/if-a-csp-experiences-a-security-incident-involving-a-hipaa-covered-entitys-or-business-associates-ephi-must-it-report-the-incident-to-the-covered-entity-or-business-associate/index.html
133. https://www.hhs.gov/hipaa/for-professionals/privacy/guidance/business-associates/index.html
134. https://www.hhs.gov/hipaa/for-professionals/covered-entities/sample-business-associate-agreement-provisions/index.html
135. https://www.hhs.gov/hipaa/for-professionals/security/laws-regulations/index.html
136. https://www.hhs.gov/sites/default/files/ocr/privacy/hipaa/administrative/securityrule/pprequirements.pdf
137. https://aws.amazon.com/health/healthcare-compliance/
138. https://aws.amazon.com/compliance/hipaa-compliance/
139. https://learn.microsoft.com/en-us/entra/identity-platform/v2-oauth2-device-code
140. https://learn.microsoft.com/en-us/entra/identity-platform/v2-app-types
141. https://learn.microsoft.com/en-us/entra/external-id/
142. https://aws.amazon.com/de/msk/faqs/
143. https://learn.microsoft.com/en-us/Entra/external-id/customers/faq-customers
144. https://learn.microsoft.com/en-us/entra/identity-platform/v2-protocols
145. https://learn.microsoft.com/it-it/Entra/identity-platform/v2-oauth2-client-creds-grant-flow

146. https://learn.microsoft.com/en-us/entra/identity-platform/v2-oauth2-client-creds-grant-flow
147. https://docs.aws.amazon.com/msk/latest/developerguide/msk-authentication.md
148. https://docs.aws.amazon.com/msk/latest/developerguide/msk-replicator-bidirectional-offset-sync.html
149. https://docs.aws.amazon.com/msk/latest/developerguide/security-iam-awsmanpol-AWSMSKReplicatorExecutionRole.html
150. https://debezium.io/documentation/reference/3.4/transformations/outbox-event-router.html
151. https://debezium.io/documentation/reference/3.7/transformations/outbox-event-router.html
152. https://docs.aws.amazon.com/msk/latest/developerguide/consumer-lag.md
153. https://aws.amazon.com/blogs/security/customize-federated-sign-in-with-new-amazon-cognito-lambda-trigger/
154. http://debezium.io/documentation/reference/integrations/outbox
155. https://debezium.io/documentation/reference/3.4/integrations/outbox.html
156. https://debezium.io/blog/2024/10/04/debezium-community-stores-with-lars-johansson/
157. https://debezium.io/documentation/reference/stable/transformations/outbox-event-router.html
158. https://debezium.io/blog/2019/02/19/reliable-microservices-data-exchange-with-the-outbox-pattern/
159. https://kafka.apache.org/25/javadoc/org/apache/kafka/clients/consumer/KafkaConsumer.html
160. https://kafka.apache.org/32/javadoc/org/apache/kafka/clients/consumer/KafkaConsumer.html
161. https://kafka.apache.org/23/javadoc/org/apache/kafka/clients/consumer/KafkaConsumer.html
162. https://kafka.apache.org/11/javadoc/org/apache/kafka/clients/consumer/KafkaConsumer.html
163. https://kafka.incubator.apache.org/0101/implementation/distribution/
164. https://www.coherehealth.com/blog/ahip-prior-authorization-commitments-health-plans
165. https://kafka.apache.org/22/javadoc/org/apache/kafka/clients/consumer/KafkaConsumer.html
166. https://cwiki.apache.org/confluence/spaces/KAFKA/pages/74683993/KIP-211+Revise+Expiration+Semantics+of+Consumer+Group+Offsets?src=contextnavchildmode
167. https://kafka.apache.org/24/javadoc/org/apache/kafka/clients/consumer/Consumer.html
168. https://docs.aws.amazon.com/jp_ja/IAM/latest/UserGuide/id_roles_providers.md
169. https://kafka.apache.org/0110/javadoc/org/apache/kafka/clients/consumer/KafkaConsumer.html
170. https://docs.aws.amazon.com/cli/latest/reference/iam/create-open-id-connect-provider.html
171. https://docs.aws.amazon.com/msk/latest/developerguide/create-cluster-console.html
172. https://docs.aws.amazon.com/msk/latest/developerguide/msk-password.html
173. https://factorhouse.io/resources/kafka/tools/kafka-ui-tools-amazon-msk/
174. https://docs.amazonaws.cn/en_us/msk/latest/developerguide/iam-access-control.html
175. https://docs.aws.amazon.com/msk/latest/developerguide/mvpc-cluster-owner-action-turn-on.html
176. https://factorhouse.io/resources/kafka/tools/kafka-tools-amazon-msk-connect/
177. https://docs.aws.amazon.com/en_en/msk/latest/developerguide/create-iam-access-control-policies.html
178. https://www.automq.com/blog/amazon-msk-security-review-regulated-workloads
179. https://docs.aws.amazon.com/fr_fr/msk/latest/developerguide/logging-API-using-cloudtrail.md
180. https://docs.aws.amazon.com/en_en/msk/latest/developerguide/msk-logging.html

181. https://aws.amazon.com/about-aws/whats-new/2023/11/amazon-msk-iam-programming-languages-clusters/
182. https://aws.amazon.com/about-aws/whats-new/2023/12/amazon-msk-iam-support-programming-languages-clusters/
183. https://aws.amazon.com/managed-blockchain/sla/
184. https://d1.awsstatic.com/legal/msk-sla/Amazon_Managed_Streaming-Apache_Kafka_SLA_Japanese_2022-05-19.pdf
185. https://cohere.io/security
186. https://docs.aws.amazon.com/msk/latest/developerguide/client-access.md
187. https://aws.amazon.com/cn/kms/sla/historial-sla-1/
188. https://www.keragon.com/blog/cohere-health-alternatives
189. https://trustcenter.cohere.com/resources-details
190. https://docs.aws.amazon.com/wellarchitected/latest/reliability-pillar/availability.html
191. https://www.getprosper.ai/blog/healthcare-ai-platform-buyers-guide
192. https://docs.aws.amazon.com/en_en/msk/latest/developerguide/consumer-lag.md
193. https://d1.awsstatic.com/legal/msk-sla/Amazon_Managed_Streaming-Apache_Kafka_SLA_Indonesian_2022-05-19.pdf
194. https://aws.amazon.com/msk/faqs/?nc1=h_ls
195. https://docs.aws.amazon.com/ru_ru/msk/latest/developerguide/monitoring.html
196. https://www.amazonaws.cn/en/legal/sla/
197. https://docs.aws.amazon.com/msk/latest/developerguide/mvpc-cross-account-update-authschemes.html
198. https://d1.awsstatic.com/legal/msk-sla/Amazon_Managed_Streaming-Apache_Kafka_SLA_French_2022-05-19.pdf
199. https://d1.awsstatic.com/legal/msk-sla/Amazon_Managed_Streaming-Apache_Kafka_SLA_Portugeuse_2022-05-19.pdf
200. https://d1.awsstatic.com/legal/Amazon_Managed_Streaming_for_Apache_Kafka_MSK/Amazon.com - Managed Streaming for Apache Kafka (MSK) Service Level Agreement - October 2019_FRA.pdf
201. https://www.linkedin.com/posts/jaredcouillard_has-been-quite-the-journey-but-happy-to-report-activity-6978450056286715904-oreS
202. https://aicurated.io/tools/cohere-health/
203. https://aihealthindex.io/vendors/cohere-health
204. https://physicianaitools.com/tools/cohere-health-platform/
205. https://www.curatrix.co/partners/cohere-health
206. https://www.coherehealth.com/news/cohere-health-named-top-prior-authorization-company
207. https://coherehealth.com/blog/cohere-health-receives-hitrust-recertification
208. https://docs.aws.amazon.com/msk/latest/developerguide/mvpc-cross-account-user-action.html
209. https://www.coherehealth.com/news/cohere-health-earns-hitrust-r2-certification
210. https://learn.microsoft.com/en-us/entra/external-id/external-identities-overview?bc=/azure/active-directory-b2c/breadcrumb/toc.json&toc=/azure/active-directory-b2c/TOC.json
211. https://learn.microsoft.com/en-us/entra/architecture/external-identity-deployment-architectures
212. https://www.hhs.gov/sites/default/files/ocr/privacy/hipaa/administrative/securityrule/riskassessment.pdf
213. https://learn.microsoft.com/en-us/office365/servicedescriptions/azure-active-directory
214. https://www.hhs.gov/hipaa/for-professionals/security/guidance/cybersecurity-newsletter-january-2026/index.html
215. https://learn.microsoft.com/en-us/entra/external-id/external-identities-pricing

216. https://www.hhs.gov/sites/default/files/ocr/privacy/hipaa/understanding/special/emergency/enforcementstatement.pdf
217. https://www.hhs.gov/sites/default/files/ocr/privacy/hipaa/administrative/privacyrule/propmods.txt
218. https://www.hhs.gov/sites/default/files/model-business-associate-agreement.pdf
219. https://www.hhs.gov/hipaa/for-professionals/faq/do-the-hipaa-rules-require-csps/index.html
220. https://www.hhs.gov/hipaa/for-professionals/special-topics/emergency-preparedness/hipaa-limited-waiver-2025-phe-washington.html
221. https://www.hhs.gov/hipaa/for-professionals/privacy/guidance/disclosures-treatment-payment-health-care-operations/index.html
222. https://aspe.hhs.gov/reports/standards-privacy-individually-identifiable-health-information-regulation-text
223. https://www.hhs.gov/hipaa/for-professionals/compliance-enforcement/audit/protocol/index.html
224. https://www.hhs.gov/sites/default/files/breach-report-to-congress-2024.pdf
225. https://www.hhs.gov/sites/default/files/breach-report-to-congress-2018.pdf
226. https://debezium.io/blog/2020/10/08/debezium-community-stories-with-renato-mefi/
227. https://docs.aws.amazon.com/dms/latest/userguide/CHAP_ReleaseNotes.html
228. https://docs.aws.amazon.com/dms/latest/userguide/CHAP_Target.html
229. https://debezium.io/documentation/reference/stable/features.html
230. https://docs.aws.amazon.com/msk/latest/developerguide/aws-access-mult-vpc.html
231. https://aws.amazon.com/blogs/big-data/secure-connectivity-patterns-to-access-amazon-msk-across-aws-regions/
232. https://docs.aws.amazon.com/msk/latest/developerguide/msk-replicator-prepare-clusters.html
233. https://docs.aws.amazon.com/msk/latest/developerguide/msk-replicator-supported-clusters-versions.html
234. https://aws.amazon.com/blogs/compute/triggering-aws-lambda-function-from-a-cross-account-amazon-managed-streaming-for-apache-kafka/
235. https://docs.aws.amazon.com/msk/latest/developerguide/limits.md
236. https://stackoverflow.com/questions/79722508/offsets-retention-minutes-not-working-as-expected
237. https://www.automq.com/blog/amazon-msk-tenant-isolation-acls-iam-quotas
238. https://cloud.google.com/blog/products/compute/how-to-use-multi-vpcs-with-google-cloud-vmware-engine
239. https://pages.awscloud.com/rs/112-TZM-766/images/20191120_AWS-BlackBelt_AmazonMSK.pdf
240. https://www.hhs.gov/hipaa/for-professionals/compliance-enforcement/audit/protocol-edited/index.html
241. https://www.hhs.gov/hipaa/for-professionals/faq/minimum-necessary/index.html
242. https://www.hhs.gov/answers/minimum-necessary/index.html
243. https://www.hhs.gov/hipaa/for-professionals/faq/may-covered-entity-rely-on-a-request-from-a-business-associate/index.html
244. https://www.hhs.gov/answers/business-associates/index.html
245. https://www.hhs.gov/sites/default/files/ocr/privacy/hipaa/administrative/securityrule/securityrulepdf.pdf
246. https://www.hhs.gov/sites/default/files/ocr/privacy/hipaa/administrative/securityrule/smallprovider.pdf
247. https://www.hhs.gov/hipaa/for-professionals/special-topics/health-information-technology/cloud-computing/index.html

248. https://www.hhs.gov/hipaa/for-professionals/privacy/laws-regulations/index.html
249. https://kafka.apache.org/41/javadoc/index-all.html
250. https://docs.aws.amazon.com/msk/latest/developerguide/security_iam_service-with-iam-id-based-policies.html
251. https://kafka.apache.org/41/javadoc/org/apache/kafka/clients/consumer/package-summary.html
252. https://docs.aws.amazon.com/msk/latest/developerguide/create-iam-access-control-policies.md
253. https://docs.aws.amazon.com/msk/latest/developerguide/create-client-iam-role.md
254. https://kafka.apache.org/41/javadoc/org/apache/kafka/common/errors/UnsupportedVersionException.html
255. https://docs.aws.amazon.com/msk/latest/developerguide/msk-replicator-iam-ref.md
256. https://aws.amazon.com/msk/sla/
257. https://docs.aws.amazon.com/msk/latest/developerguide/iam-access-control-use-cases.html
258. https://kafka.apache.org/41/javadoc/org/apache/kafka/clients/consumer/KafkaConsumer.html
259. https://kafka.apache.org/41/javadoc/org/apache/kafka/clients/consumer/Consumer.html
260. https://docs.aws.amazon.com/msk/latest/developerguide/kafka-actions.html
261. https://docs.aws.amazon.com/msk/latest/developerguide/create-iam-access-control-policies.html
262. https://docs.aws.amazon.com/msk/latest/developerguide/msk-connect-service-execution-role.html
263. https://docs.aws.amazon.com/msk/latest/developerguide/msk-replicator-perms-topic-replication.html
264. https://docs.aws.amazon.com/msk/latest/developerguide/iam-access-control.html
265. https://docs.aws.amazon.com/msk/latest/developerguide/kafka_apis_iam.html
266. https://docs.aws.amazon.com/msk/latest/developerguide/kafka_apis_iam.md
267. https://docs.aws.amazon.com/msk/latest/developerguide/configure-clients-for-iam-access-control.html
268. https://docs.aws.amazon.com/msk/latest/developerguide/msk-replicator-supported-configs.html
269. https://docs.aws.amazon.com/msk/latest/developerguide/serverless.md
270. https://docs.aws.amazon.com/msk/latest/developerguide/msk-configuration-properties.md
271. https://docs.aws.amazon.com/msk/latest/developerguide/monitoring.md
272. https://docs.aws.amazon.com/msk/latest/developerguide/troubleshooting.md
273. https://docs.aws.amazon.com/msk/latest/developerguide/msk-data-delivery-iceberg-cloudtrail.html
274. https://docs.aws.amazon.com/msk/latest/developerguide/msk-acls.md
275. https://docs.aws.amazon.com/msk/latest/developerguide/bestpractices-cw-alarms.md
276. https://docs.aws.amazon.com/msk/1.0/apireference/clusters-clusterarn-monitoring.md
277. https://docs.aws.amazon.com/msk/latest/developerguide/msk-replicator-troubleshooting.md
278. https://learn.microsoft.com/en-us/Entra/external-id/customers/overview-customers-ciam
279. https://learn.microsoft.com/en-us/entra/external-id/customers/concept-planning-your-solution
280. https://learn.microsoft.com/ro-ro/Entra/external-id/tenant-configurations
281. https://learn.microsoft.com/en-us/entra/external-id/tenant-configurations
282. https://learn.microsoft.com/en-us/azure/active-directory/external-identities/external-identities-overview
283. https://learn.microsoft.com/en-us/training/modules/manage-identities-microsoft-entra-id/7-secure-external-microsoft-entra-identities

284. https://learn.microsoft.com/en-us/training/modules/manage-security-controls-identity-access/8-secure-external-microsoft-entra-identities/
285. https://learn.microsoft.com/en-us/entra/external-id/external-identities-overview
286. https://learn.microsoft.com/en-us/entra/external-id/customers/faq-customers
287. https://learn.microsoft.com/en-us/entra/external-id/customers/concept-supported-features-customers
288. https://learn.microsoft.com/en-us/answers/questions/5855639/sso-customer-entra-id-as-idp-for-entra-external-id
289. https://learn.microsoft.com/en-us/answers/questions/5858661/sso-b2b-approach-for-customer-entra-id-as-idp-for
290. https://docs.aws.amazon.com/msk/latest/developerguide/msk-authentication.html
291. https://learn.microsoft.com/en-us/answers/questions/5804791/how-to-allow-microsoft-work-personal-accounts-in-e
292. https://learn.microsoft.com/en-us/entra/external-id/what-is-b2b
293. https://docs.aws.amazon.com/msk/latest/developerguide/how-to-use-iam-access-control.md
294. https://aws.amazon.com/msk/sla/historical/
295. https://docs.aws.amazon.com/msk/latest/developerguide/how-to-use-iam-access-control.html
296. https://docs.aws.amazon.com/msk/latest/developerguide/create-iam-access-control-cluster-in-console.html
297. https://docs.aws.amazon.com/msk/latest/developerguide/security-iam.html
298. https://docs.aws.amazon.com/msk/latest/developerguide/security_iam_service-with-iam.html
299. https://docs.aws.amazon.com/msk/latest/developerguide/serverless.html
300. https://docs.aws.amazon.com/msk/latest/developerguide/msk-replicator-ser-additional-perms.html
301. https://docs.aws.amazon.com/msk/latest/developerguide/msk-replicator-external-prereqs.html
302. https://docs.aws.amazon.com/msk/latest/developerguide/iam-access-control-use-cases.md
303. https://www.coherehealth.com/news/cohere-addresses-prior-authorization-compliance-gaps
304. https://www.coherehealth.com/news/cohere-vori-partnership
305. https://www.coherehealth.com/news/cms-electronic-prior-authorization-acceleration-initiative-support
306. https://www.coherehealth.com/blog/healthcare-interoperability-cms-ahip-requirements
307. https://www.coherehealth.com/solutions/health-plan
308. https://www.coherehealth.com/blog/dtr-workflows-medical-policy-digitization